



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

DDA2001: Introduction to Data Science

Lecture 6: Common Distributions (Discrete)

Shuang Li

Recap: 1 - Some Useful formulas

Formula 1

$$\text{Linearity: } E[\sum_i X_i] = \sum_i E[X_i]$$



No assumption on X_i

A More General Formula

$$E[\sum_i C_i X_i] = \sum_i C_i E[X_i]$$



C_i is a constant

How to use the linearity of expectation?

Want to calculate $E[X]$, where X is a 'complicated' random variable

- Step 1: Figure out a suitable decomposition for X such that $X = \sum_i X_i$
- Step 2: Compute $E[X_i]$
- Step 3: Use the linearity of expectation ($E[X] = \sum_i E[X_i]$)

Formula 2

$$E[g(X)] = \sum_x g(x)P(X = x) = \sum_x g(x)f(x)$$

- Note: $E[g(X)] \neq g(E[X])$

- $g(X)$ is also a random variable
- By definition of expectation, you need to figure out the pmf for $g(X)$ first

- Toss a coin: head as 1, tail as -1.
- Then $E[X^2] = 1^2 \times \frac{1}{2} + (-1)^2 \times \frac{1}{2} = 1$
- But $(E[X])^2 = 0$

$$g(x) = x^2$$

Variance

$$E[(X - E[X])^2]$$

$$X^2 - 2XE[X] + (E[X])^2$$



- $E[C] = C$ for constant C
- $E[X]$ is a constant
- $E[E[X]] = E[X]$

$$E[(X - E[X])^2] = E[X^2] - 2E[X]E[X] + (E[X])^2$$

$$- (E[X])^2$$

More useful

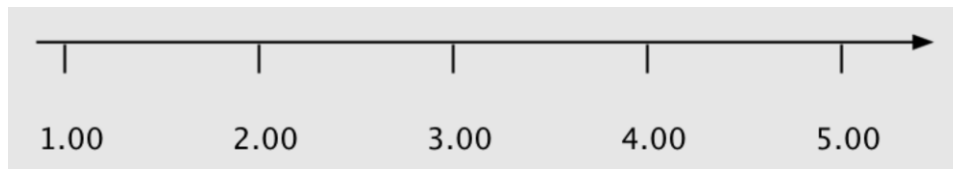
$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

- Could the linearity properties for expectation be extended to variance?
- Suppose $X_1, X_2, \dots$ are independent, we can have
$$\begin{aligned} & Var(X_1 + X_2 + X_3 + \dots) \\ &= Var(X_1) + Var(X_2) + Var(X_3) + \dots \end{aligned}$$

2. Common Distribution (Discrete)

Example 1

- You are the owner of a blind box store.
- Suppose that during each minute of the day, there is a probability p that a customer shows up in your store, and a probability $1 - p$ that no one shows up. Suppose at most one customer can show each minute.



- Let X_i be the number of customers arrive in the i -th minute. What is the distribution of X_i ?
- Let $Y_N = X_1 + \dots + X_N$ be the number of customers arrive in the first N minutes of the day. What is the distribution of Y_N ?
- Let Z be the minutes taken for the first customer to show up. What is the distribution of Z ?

During each minute of the day, there is a probability p that a customer shows up in your store, and a probability $1 - p$ that no one shows up.

?	?	?
Whether a customer arrives during a specific minute.	The number of customers arrive in N minutes	At which minute does the first customer arrive

1. Bernoulli distribution

- Take value 1 with probability p and value 0 with probability $1 - p$.

1. Bernoulli distribution

- Take value 1 with probability p and value 0 with probability $1 - p$.
- $f(1) = p, f(0) = 1 - p$

1. Bernoulli distribution

- Take value 1 with probability p and value 0 with probability $1 - p$.
- $f(1) = p, f(0) = 1 - p$

Other Examples

- Toss a coin
- Win or lose in a game



1. Bernoulli distribution

- Take value 1 with probability p and value 0 with probability $1 - p$.
- $f(1) = p, f(0) = 1 - p$

Other Examples

- Toss a coin
- Win or lose in a game

Exercise: What is $E[X_i]$ and $\text{Var}(X_i)$?



1. Bernoulli distribution

- Take value 1 with probability p and value 0 with probability $1 - p$.
- $f(1) = p, f(0) = 1 - p$

Other Examples

- Toss a coin
- Win or lose in a game

Exercise: What is $E[X_i]$ and $\text{Var}(X_i)$?

- Mean= p
- Variance= $p(1-p)$



During each minute of the day, there is a probability p that a customer shows up in your store, and a probability $1 - p$ that no one shows up.

Bernoulli Distribution	?	?
Whether a customer arrives during a specific minute.	The number of customers arrive in N minutes	At which minute does the first customer arrive

2. Binomial distribution

- One Bernoulli trial: take value 1 with probability p and value 0 with probability $1 - p$.
- N Bernoulli trials??

2. Binomial distribution

- One Bernoulli trial: take value 1 with probability p and value 0 with probability $1 - p$.
- N Bernoulli trials?? -> Binomial distribution

2. Binomial distribution

- One Bernoulli trial: take value 1 with probability p and value 0 with probability $1 - p$.
- N Bernoulli trials?? -> Binomial distribution
- With parameters N and p
- N independent experiments
- Each experiment: success (with probability p) or failure (with probability $1 - p$).
- X : the number of success.

2. Binomial distribution

- With parameters N and p
- N independent experiments
- Each experiment: success (with probability p) or failure (with probability $1 - p$).
- X : the number of success.

$$\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \qquad \binom{n}{k} = \frac{n!}{k!(n - k)!}$$

2. Binomial distribution

- With parameters N and p
- N independent experiments
- Each experiment: success (with probability p) or failure (with probability $1 - p$).
- X : the number of success.

$$\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}$$

- Mean = np
- Variance = $np(1-p)$

During each minute of the day, there is a probability p that a customer shows up in your store, and a probability $1 - p$ that no one shows up.

Bernoulli Distribution	Binomial Distribution	?
Whether a customer arrives during a specific minute.	The number of customers arrive in N minutes	At which minute does the first customer arrive

2. Binomial distribution: Application

- A buyer buys N products from a producer.



2. Binomial distribution: Application

- A buyer buys N products from a producer.
- Each product can be flawed (w.p. p) or not (w.p. $1-p$)



2. Binomial distribution: Application

- A buyer buys N products from a producer.
- Each product can be flawed (w.p. p) or not (w.p. $1-p$)
- If k products are flawed, the producer has to refund R_k



2. Binomial distribution: Application

- A buyer buys N products from a producer.
- Each product can be flawed (w.p. p) or not (w.p. $1-p$)
- If k products are flawed, the producer has to refund R_k
- How much the producer will pay to the buyer on average?



2. Binomial distribution: Application

- A buyer buys N products from a producer.
- Each product can be flawed (w.p. p) or not (w.p. $1-p$)
- If k products are flawed, the producer has to refund R_k
- How much the producer will pay to the buyer on average?

$$R(N,p) = \sum_{k=1}^N \Pr(X = k) R_k$$



2. Binomial distribution: Application

- Is increasing production yield rate always beneficial for a factory?
- Production Yield Rate: the percentage of non-defective items of all produced items



2. Binomial distribution: Application

- Suppose the producer can incur cost $C(p)$ to ensure that the product is flawed w.p. p .



2. Binomial distribution: Application

- Suppose the producer can incur cost $C(p)$ to ensure that the product is flawed w.p. p .
- How much to invest?



2. Binomial distribution: Application

- Suppose the producer can incur cost $C(p)$ to ensure that the product is flawed w.p. p .
- How much to invest?
- Choose p such that $R(N,p) + C(p)$ is minimized



3.Geometric distribution

- Continuously draw a Bernoulli R.V.
- The X-th draw is the first success.
- X follows a geometric distribution.

$$\Pr(X = k) = (1 - p)^{k-1}p$$

3.Geometric distribution

- Continuously draw a Bernoulli R.V.
- The X-th draw is the first success.
- X follows a geometric distribution.

$$\Pr(X = k) = (1 - p)^{k-1}p$$

- Mean= $1/p$
- Variance= $(1-p)/p^2$

3.Geometric distribution

- Continuously draw a Bernoulli R.V.
- The X -th draw is the first success.
- X follows a geometric distribution.

$$\Pr(X = k) = (1 - p)^{k-1}p$$

Example

- Toss a coin until there is a head.
- Buy a blind box until you get a Harry Potter

During each minute of the day, there is a probability p that a customer shows up in your store, and a probability $1 - p$ that no one shows up.

Bernoulli Distribution	Binomial Distribution	Geometric Distribution
Whether a customer arrives during a specific minute.	The number of customers arrive in N minutes	At which minute does the first customer arrive

3.Geometric distribution: Application

- At each day, a machine breaks down with a probability p .



3. Geometric distribution: Application

- At each day, a machine breaks down with a probability p .
- What's the expected duration before the machine breaks down?



Connections

- A machine breaks down w.p. p on each date.

Bernoulli distribution	Binomial distribution	Geometric Distribution
Whether the machine breaks down on a specific date.		

Connections

- A machine breaks down w.p. p on each date.

Bernoulli distribution	Binomial distribution	Geometric Distribution
Whether the machine breaks down on a specific date.	The number of breakdowns during the first N dates.	

Connections

- A machine breaks down w.p. p on each date.

Bernoulli distribution	Binomial distribution	Geometric Distribution
Whether the machine breaks down on a specific date.	The number of breakdowns during the first N dates.	The first date the machine breaks down